EXATHON INDIA 2026 · TRACK 04 · THE WORK BEHIND THE WORK

TrustLayer

Stops broken data before it reaches a dashboard, shows who was protected, and explains what happened.

Sharvesh S · Built on Exasol Personal · github.com/SHARVESH-005/trustlayer

THE PROBLEM

Pipelines fail quietly. Dashboards show the damage.

2 AM

A job half-fails

A retry duplicates rows, a column goes missing, amounts switch to cents. Nothing errors, so nothing alerts.

9 AM

The CFO finds it

Wrong numbers on the executive dashboard. Nobody can say which reports are affected, or since when.

After

Trust is gone

Teams re-check reports by hand and stop relying on data. Alerts after the fact don’t undo the damage.

Who has it: data engineers on call, analytics teams, and every business owner who reads a dashboard.

THE SOLUTION

A quarantine gate in front of every dashboard

Hourly batch
lands in STAGE

→

12 checks in Exasol
vs learned baselines

→

Pass: CURATED
dashboards read only this

Fail: QUARANTINE

The batch is held, an incident opens, and dashboards keep the last good data.

Then explain and fix

Blast radius from Exasol’s dependency catalog, a Gemini report, and one-click retry.

HOW THE CHECKS WORK

Checks learn what normal looks like

Hour-of-week baselines

Monday 9 AM is compared with past Mondays at 9 AM, over the last six weeks.

Robust stats, inside Exasol

A Python UDF computes median and MAD, so one outage in the history barely moves the baseline.

Adapts to drift

Expected volume follows recent traffic; quality rates compare by hour of day over two weeks.

Two kinds of evidence

An alert needs a change that is statistically unusual **and** large enough to matter.

THE DEMO

One failure, start to finish

1

Break it

Upstream switches amounts from dollars to cents.

2

Blocked

Average fare $26 → $2,600+. Quarantined after 12 checks in about 1 s.

3

Protected

4 dashboards and owners named from EXA_DBA_DEPENDENCIES.

4

Explained

Gemini: “a unit conversion error, cents instead of dollars”.

5

Fixed

Retry re-runs the hour, the incident closes, trust is back to 100.

The trust score is honest: blocking keeps dashboards **correct** but **behind**, so it drops until the hour is re-run.

AI INCIDENT REPORTS

Gemini finds the cause without being told

What we injected	What Gemini concluded from the evidence
Amounts switched to cents	“a unit conversion error (e.g., cents instead of dollars)”, high confidence
Tip column emptied	“the column was dropped or not populated” upstream
Retried job re-sent rows	“likely a double ingestion or a failure in the deduplication step”
Fare signs flipped	traced 41.8% invalid rows to 2,173 negative totals out of 5,201

Gemini sees only aggregates computed in Exasol, never raw rows. One call per incident, cached; model fallback; a template report when offline.

WHY EXASOL

Checks run where the data lives

9.55M
real NYC taxi trips landed, Jan–Mar 2024

<1 min
to publish 6.3M rows of history

~20 s
to learn every baseline with a Python UDF

~1 s
for 12 checks on an hourly batch

Baselines, checks, quarantine and lineage all use Exasol itself: SQL, an in-database Python UDF and the EXA_DBA_DEPENDENCIES catalog. No data is copied out.

RESULTS ON REAL DATA

Quiet when it should be, loud when it must be

0

false quarantines across 96 consecutive real hourly batches

7 / 7

failure types blocked: duplicates, volume drop, empty load, null spike, dropped column, bad values, unit change

1

real upstream drift surfaced: missing passenger counts rose from ~5% to 20–28% on March 2024 evenings

TRACK 04 FIT

The work behind the work

Track pain point	TrustLayer	Status
Pipelines that fail overnight without warning	Learned checks on every batch, incidents opened automatically	**Built**
Dashboards the business no longer trusts	Bad batches blocked; live trust score and embeddable badge	**Built**
Data requests that queue for weeks	Plain-English incident reports; ask questions in English next	Partly built
Platform cost growing faster than usage	Waste report from Exasol’s own statistics	Roadmap

WHAT'S NEXT

The reason to believe your dashboard

Roadmap

- Ask questions in plain English, answered with SQL on Exasol
- Waste report from Exasol's statistics tables
- Slack and Teams alerts; auto-fix for duplicates
- Any table, not just taxi trips: config-driven checks

Try it

```
github.com/SHARVESH-005/trustlayer
```

```
python demo.py
```

Demo video (3 min): watch on Google Drive

Thank you · Sharvesh S